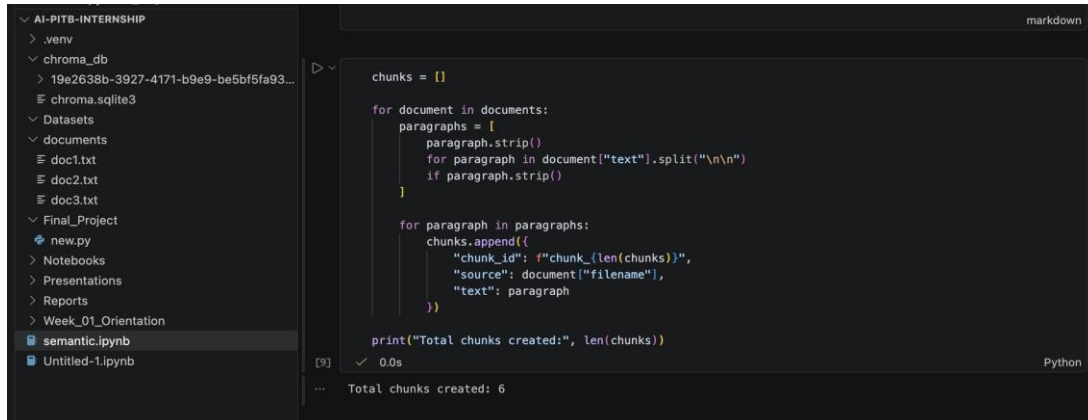


Semantic Search System

CHUNKS STORAGE COUNT



The screenshot shows a Jupyter Notebook interface. The left sidebar displays a file explorer with a project structure including folders like 'AI-PITB-INTERNSHIP', 'chroma_db', 'Datasets', 'documents', 'Final_Project', and 'Notebooks'. The 'documents' folder is expanded, showing 'doc1.txt', 'doc2.txt', and 'doc3.txt'. The main notebook area shows a Python script that iterates through documents, splits them into paragraphs, and stores each paragraph as a chunk in a list. The script prints the total number of chunks created, which is 6.

```
chunks = []

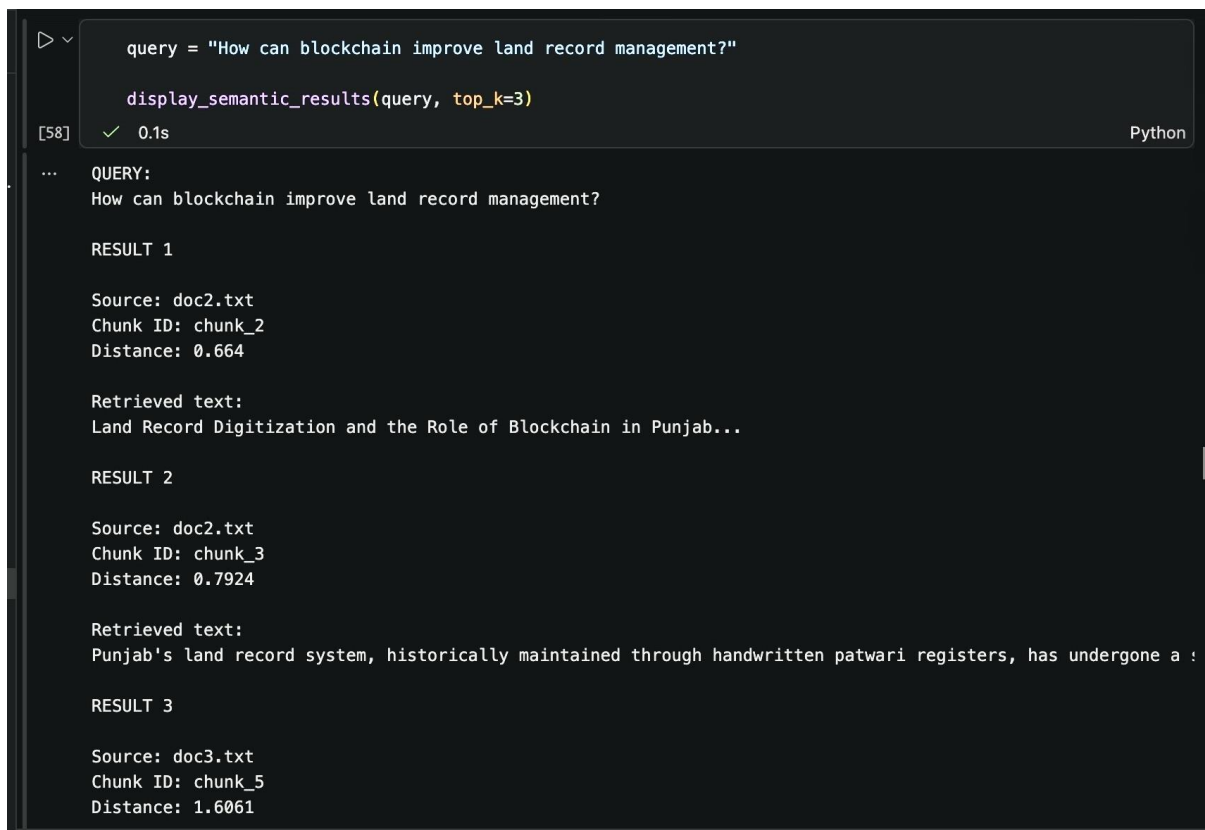
for document in documents:
    paragraphs = [
        paragraph.strip()
        for paragraph in document["text"].split("\n\n")
        if paragraph.strip()
    ]

    for paragraph in paragraphs:
        chunks.append({
            "chunk_id": f"chunk_{len(chunks)}",
            "source": document["filename"],
            "text": paragraph
        })

print("Total chunks created:", len(chunks))
```

Output: Total chunks created: 6

SEMENTIC SEARCH RESULT



The screenshot shows a Jupyter Notebook interface. The main notebook area shows a Python script that queries the semantic search system for the query "How can blockchain improve land record management?". The script displays the top 3 results. The output shows the query, the results, and the retrieved text for each result.

```
query = "How can blockchain improve land record management?"

display_semantic_results(query, top_k=3)
```

Output:

```
[58] ✓ 0.1s Python

... QUERY:
How can blockchain improve land record management?

RESULT 1

Source: doc2.txt
Chunk ID: chunk_2
Distance: 0.664

Retrieved text:
Land Record Digitization and the Role of Blockchain in Punjab...

RESULT 2

Source: doc2.txt
Chunk ID: chunk_3
Distance: 0.7924

Retrieved text:
Punjab's land record system, historically maintained through handwritten patwari registers, has undergone a :

RESULT 3

Source: doc3.txt
Chunk ID: chunk_5
Distance: 1.6061
```

COMPARISON SEMANTIC VS KEYWORD SEARCH

```
comparison_queries = [
    "How can blockchain make land records more secure?",
    "How can AI improve public sector services?",
    "How can digital technology help citizens access government services?"
]
for query in comparison_queries:

    print()
    print("QUERY:", query)
    print()

    print("\nKEYWORD SEARCH")
    print()

    display_keyword_results(
        query,
        top_k=2
    )

    print("\nSEMANTIC SEARCH")
    print()

    display_semantic_results(
        query,
        top_k=2
    )
```

[66] ✓ 0.2s Python

RESULTS

```
...
QUERY: How can blockchain make land records more secure?

KEYWORD SEARCH

QUERY:
How can blockchain make land records more secure?

RESULT 1

Keyword score: 4
Source: doc2.txt
Chunk ID: chunk_3

Retrieved text:
Punjab's land record system, historically maintained through handwritten patwari registers, has undergone a :

RESULT 2

Keyword score: 2
Source: doc2.txt
Chunk ID: chunk_2

Retrieved text:
Land Record Digitization and the Role of Blockchain in Punjab

SEMANTIC SEARCH
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
... SEMANTIC SEARCH

QUERY:
How can blockchain make land records more secure?

RESULT 1
Source: doc2.txt
Chunk ID: chunk_2
Distance: 0.7326

Retrieved text:
Land Record Digitization and the Role of Blockchain in Punjab...

RESULT 2
Source: doc2.txt
Chunk ID: chunk_3
Distance: 0.838

Retrieved text:
Punjab's land record system, historically maintained through handwritten patwari registers, has undergone a

QUERY: How can AI improve public sector services?

KEYWORD SEARCH

QUERY:
How can AI improve public sector services?
```

```
... KEYWORD SEARCH

QUERY:
How can AI improve public sector services?

RESULT 1

Keyword score: 2
Source: doc3.txt
Chunk ID: chunk_4

Retrieved text:
Artificial Intelligence Adoption in Pakistan's Public Sector

RESULT 2

Keyword score: 2
Source: doc3.txt
Chunk ID: chunk_5

Retrieved text:
Government bodies in Pakistan, including provincial IT boards, have begun exploring generative AI tools to i

SEMANTIC SEARCH

QUERY:
How can AI improve public sector services?

RESULT 1
```

```
... SEMANTIC SEARCH

QUERY:
How can AI improve public sector services?

RESULT 1
Source: doc3.txt
Chunk ID: chunk_4
Distance: 0.9639

Retrieved text:
Artificial Intelligence Adoption in Pakistan's Public Sector...

RESULT 2
Source: doc3.txt
Chunk ID: chunk_5
Distance: 1.1003

Retrieved text:
Government bodies in Pakistan, including provincial IT boards, have begun exploring generative AI tools to i

QUERY: How can digital technology help citizens access government services?

KEYWORD SEARCH

QUERY:
How can digital technology help citizens access government services?
```

```
... RESULT 2

Keyword score: 4
Source: doc2.txt
Chunk ID: chunk_3

Retrieved text:
Punjab's land record system, historically maintained through handwritten patwari registers, has undergone a :

SEMANTIC SEARCH

QUERY:
How can digital technology help citizens access government services?

RESULT 1
Source: doc1.txt
Chunk ID: chunk_0
Distance: 1.0787

Retrieved text:
Digital Service Delivery in Punjab: E-Rozgaar and Citizen Facilitation Centers...

RESULT 2
Source: doc3.txt
Chunk ID: chunk_5
Distance: 1.0991

Retrieved text:
Government bodies in Pakistan, including provincial IT boards, have begun exploring generative AI tools to ir
```

